Calli-VA: A Visual Analytics System for Analyzing and Comparing Chinese Calligraphic Styles

Jincheng Li, Jinpeng Wu, Shaocong Tan, Lin Du, Yu Zhang, Chaofan Yang, Jiadi Zhang, Rebecca Ruige Xu, Rui Shi, Lu Bai, and Xiaoru Yuan

Abstract—Chinese calligraphy is a quintessential element of Chinese cultural heritage. Analyzing and comparing calligraphic styles not only enhances the appreciation, learning, and advancement of calligraphy but also provides valuable insights into ancient China. However, such analysis remains challenging due to the limited scalability and possible inconsistencies of qualitative methods, as well as usability and misalignment issues in conventional quantitative approaches. We propose Calli-VA, a visual analytics system, to address these challenges. Calli-VA extracts character images and their corresponding strokes from original works and characterizes each character using systematic criteria. During analysis, the system defines the analysis scope by overview and uncovers relationships between characters. Explanation and recommendation mechanisms are integrated to help users understand patterns and guide further exploration. A documentation feature allows users to record and share their findings. We demonstrate the effectiveness of Calli-VA through three case studies and expert feedback.

Index Terms—Chinese calligraphy style analysis, image analysis, digital humanities, visual analytics.

1 Introduction

Chinese calligraphy, a quintessential element of Chinese cultural heritage, blends the functional purpose of writing with profound aesthetic expression. Over thousands of years, numerous renowned calligraphers and notable works have emerged. The same calligrapher may exhibit varying styles at different stages, and styles also differ between calligraphers. These styles often show relationships of imitation, inheritance, and development. By analyzing and comparing calligraphic styles, we can better appreciate calligraphy's aesthetic qualities, identify stylistic signatures, and facilitate learning and development of calligraphy. Such analysis also offers a lens for exploring broader shifts in Chinese philosophy [51], art [26], politics [54], and more.

We have worked closely with several calligraphy scholars to enhance the effectiveness of calligraphy style analysis. Through preliminary interviews and research, we identified their current practices and potential new approaches. Typically, scholars conduct qualitative analysis by observing characters and drawing conclusions from these observations. The characters are manually selected, often focusing on commonly used ones. Traditional calligraphy theories, such as balance, variety, continuity, and contrast [5], serve as foundational guidance for analysis. Systematic criteria [38] composed of multiple theories have been well established, enabling more comprehensive analysis.

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Advancements in image processing have led to the development of quantitative methods [11, 24, 29, 41, 42, 48, 53, 64]. These methods typically define features based on the theories mentioned earlier, with some further generating scores. Moreover, Zhang et al. [62] introduced a visual analysis system to help the public understand and appreciate calligraphic styles, which also relies on defined features.

Through discussions with collaborating calligraphy scholars, we identified deficiencies in both qualitative and quantitative methods. The main limitation of qualitative analysis is its lack of scalability and inherent inconsistency. This method requires careful observation and judgment of each character, making it highly time-consuming. As a result, analyzing large collections of works becomes nearly impossible, limiting comprehensive understanding of calligraphic styles. In addition, qualitative analysis heavily relies on personal knowledge and experience, making it inherently subjective. Although this subjectivity can bring valuable diverse perspectives, it sometimes results in inconsistencies when similar methodologies yield conflicting conclusions.

While calligraphy scholars are open to quantitative methods, these methods remain underutilized due to usability and misalignment issues. Usability is limited because such methods often overlook scholars' limited expertise in image processing and data analysis. Typically, they require individual character images as input, forcing scholars to perform tedious preprocessing before analysis can even begin. Once feature values are extracted, scholars often struggle to derive meaningful insights from the data. Notably, score-based methods are unsuitable, as scholars study works by renowned calligraphers and aim to appreciate their styles rather than judge them as "good" or "bad." The misalignment issue lies in that although these methods are based on calligraphy theories, they are sometimes incomplete due to insufficient integration with systematic criteria. Additionally, existing visual analysis method [62] is for public use and lacks the depth needed for research purposes.

To overcome these limitations, we propose Calli-VA, a visual analytics system that provides calligraphy scholars with full-process support for analyzing and comparing Chinese calligraphic styles. The system begins with semi-automatic preprocessing to extract character images and their corresponding strokes, followed by character characterization based on high-level, systematic criteria. During the analysis phase, Calli-VA starts with the overview to define the analysis scope and proceeding to detailed analysis. The system incorporates explanation and recommendation mechanisms to enhance users' understanding of patterns and enable effective adjustment of analysis granularities and perspectives. Finally, the system includes a documentation feature to record their analysis results for further study and communication.

The main contributions of this paper include the following:

- Collaborating with calligraphy scholars, we identified the challenges

in current practices and quantitative approaches for calligraphy style analysis and derived design requirements.
- We proposed Calli-VA, a visual analytics system to help scholars analyze and compare calligraphic styles more effectively.

2 Related Work

We review the literature on computer-aided Chinese calligraphy analysis, visual analysis of images, and visual analytics for humanities.

2.1 Computer-Aided Calligraphy Analysis

Chinese calligraphy analysis has received growing attention in the computer field. Here we focus on areas relevant to our work, including character recognition, stroke extraction, and aesthetic evaluation.

Chinese character recognition has been a long-standing research focus. Traditional methods include template matching [4], blurring [17], and statistical structure modeling [18]. However, recognition accuracy remains limited due to challenges such as complex structures and variations in writing styles [6]. Recently, deep learning approaches [46, 61] have helped address these challenges. In this work, we use the state-of-the-art method PP-OCR [8], which also supports text detection, enabling the extraction of character bounding boxes.

Stroke extraction refers to extracting every individual stroke of a character. Early methods relied on image morphological features such as skeletonization [16]. Advanced deep learning approaches [32, 45] addressed challenges like stroke intersections and variability. Li et al. [25] combined image registration with semantic segmentation, enhancing the precision of stroke alignment and extraction. We adopt Li et al.'s model for its open-source availability and high accuracy, supplementing it with manual adjustments to ensure greater reliability.

As discussed in Sec. 1, aesthetic evaluation methods can be categorized into two types: those that directly use manually defined features [11, 24, 29, 48, 64] and those that incorporate machine learning [41, 42, 53]. While the latter also rely on manually defined features, they train models using human-provided aesthetic scores, enabling results that more closely reflect human perception. However, the features used in these methods are typically low-level and lack systematic depth. Moreover, machine learning-based methods usually produce a single score that lacks detailed insights and interpretability. Therefore, we opt to manually define features based on higher-level, systematic criteria.

2.2 Visual Analysis of Images

Characters are represented as image collections. Visual analysis of such collections typically involves converting images into a feature space and visualizing their mutual similarities. Image collections are often accompanied by metadata, such as keywords or descriptions, which can be used to group images based on semantic similarity [55]. When textual metadata is unavailable, techniques like semantic extraction [56] can automatically identify image content. Moreover, visual similarity can be assessed using either manually defined features [15, 47] or deep learning models [44, 58]. Images can then be displayed using similarity-based techniques such as projections [34, 55] or organized into relational forms like cluster hierarchies [20] and semantic networks [13].

In our case, characters typically have limited metadata. Given the limited interpretability of deep learning models and the availability of well-established calligraphy aesthetic criteria, we opt to manually define features for similarity calculation. For visualization, we use the projection to clearly illustrate relationships between characters.

2.3 Visual Analytics for Humanities

Digital humanities is an interdisciplinary field that integrates computational technology into the humanities [36]. A notable direction of collaboration is with visual analytics [28, 49], where various methods have been proposed to support humanists in diverse tasks. Recent examples include catalog annotation [39], tracing painting provenance [59], analyzing pottery motif evolution [22], studying book circulation [14, 52], and performing cohort analysis [60]. Moreover, Xing et al. [52] described a design study and adapted the core phase of the nine-stage framework [37] into iterative design cycles.

For calligraphy style analysis, Zhang et al. [62] introduced a visual analysis method. However, it does not incorporate stroke-level features, limiting users' ability to conduct more detailed analysis. Besides, its visual design offers limited support for key tasks such as selecting characters, understanding patterns, shifting analytical focus, and recording results. In contrast, our approach is grounded in a thorough investigation of user needs, offering comprehensive character characterization and sufficient support throughout the analysis process.

3 Overview

In the early stages, we conducted multiple rounds of interviews and discussions with collaborating calligraphy scholars. This process aimed to deepen our understanding of their current practices and other promising approaches, introduce them to the potential of visual analytics, and build consensus on the system's design. Details on current practices, alternative approaches, and their limitations are provided in Sec. 1. This section introduces the data, analysis scope, and design requirements.

3.1 Data Description and Analysis Scope

The dataset contains 17 Tang dynasty inscriptions by 12 calligraphers, including raw text, images, and temporal information. Of these, 15 belong to the Wang Xizhi Semi-cursive Script Series, featuring works by calligraphers heavily influenced by Wang's style. The remaining two are from the Collected Characters of Wang Xizhi Series, composed of characters sourced from Wang's original calligraphy. There are 17,835 character instances. Wang Xizhi, known as the "Sage of Calligraphy," profoundly influenced the development of Chinese calligraphy.

Current practices show that analysis is conducted at the character level, with the scope varying along two dimensions: the number of objects (calligraphers or works) involved and the number of characters analyzed. Different scopes yield different insights. Analyzing characters within a single object focuses on identifying its stylistic characteristics, while analyzing characters across multiple objects highlights stylistic similarities and differences. Additionally, single-character analysis explores stylistic variation across different instances, whereas multi-character analysis emphasizes comparisons between characters.

3.2 Design Requirements

First, our system must support the various analysis scopes required by scholars. Interviews and discussions also revealed the need to assist scholars before and after analysis, specifically in selecting characters and recording findings. We summarize design requirements as follows.

R1: Offer entry points for character analysis. When entering the system, users need to select characters to analyze. Given the large number of characters, we follow the information-seeking mantra: "overview first, zoom and filter, details on demand" [40], providing overviews to support efficient character selection.

- **R1.1: Provide an overview of works and calligraphers.** Characters appear in works written by calligraphers. The system should display information about works and their calligraphers while supporting browsing of individual works. This helps users form an initial impression and narrow down their selections.
- **R1.2: Provide an overview of characters.** The system should enable users to view original characters for intuitive selection. Meanwhile, it must incorporate effective character aggregation strategies to prevent users from feeling overwhelmed.

R2: Allow flexible and comprehensive character analysis. Since multiple analysis scopes exist, the system should offer strong support for each and allow users to switch between them smoothly.

- **R2.1: Characterize characters from multiple perspectives.** The system should comprehensively characterize characters using the perspectives defined by established, systematic criteria. Results should be presented intuitively to provide a detailed foundation for analysis. The system should also extract and display relationships between characters to aid in pattern identification. Furthermore, users should be able to focus on the perspectives most relevant to their analysis, reducing distractions from unrelated features.

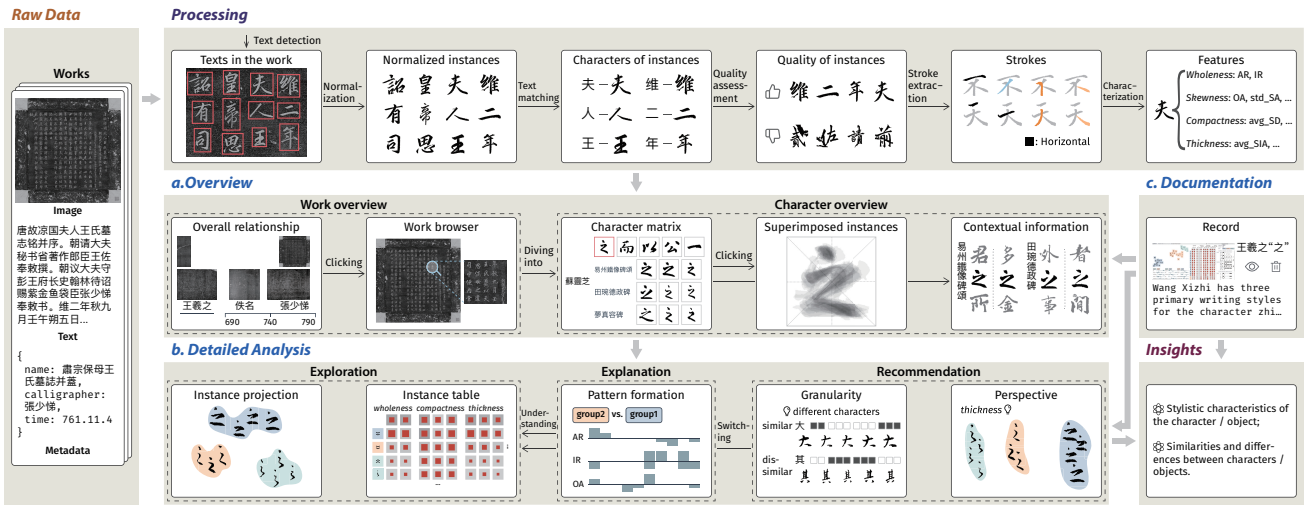


Fig. 1: The calligraphy style analysis workflow. Original calligraphy works serve as input. During processing, character images and strokes are extracted, followed by character characterization based on systematic criteria. The visual analysis workflow comprises three components: overview, detailed analysis, and documentation, helping users identify characters for analysis, conduct character analysis, and record findings, respectively.

- **R2.2: Explore characters across multiple granularities.** Users can perform single- or multi-character analyses on individual or multiple objects and further refine their focus by selecting subsets of instances to form groups. The system must support users' needs across all granularities while keeping characters well-organized and easily traceable to ensure clarity and avoid confusion.
- **R2.3: Integrate pattern explanation mechanisms.** When a pattern emerges, the system should explain its formation by highlighting the most influential features, helping users deepen their understanding.
- **R2.4: Facilitate adjustments to granularities and perspectives.** Users may adopt diverse analysis granularities and perspectives. When analysis is exploratory rather than goal-driven, trial-and-error approaches can be inefficient. To alleviate this, the system should offer helpful recommendations, such as suggesting perspectives and granularities that reveal significant patterns.

R3: Support user documentation. The system should support summarizing insights to complete the analytical loop and facilitate sharing. Since users may switch between different analysis granularities and perspectives, and each step may yield new insights, the system should allow them to save intermediate results and provide necessary auxiliary information for better documentation.

4 Data Modeling

This section outlines the data processing procedures, including character and stroke extraction and character characterization (Fig. 1).

4.1 Character and Stroke Extraction

This part primarily involves character extraction, character quality assessment, and stroke extraction.

4.1.1 Character Extraction

Character extraction involves obtaining character instance images from original works and matching them to corresponding characters. It has three steps: text detection, normalization, and text matching.

Text detection identifies the bounding box of each character instance. We use PP-OCR [8] for automatic detection and supplement it with manual annotation to add missing bounding boxes and adjust existing ones to fully enclose the characters.

Normalization starts by cropping each character instance image using its bounding box, then binarizing it and flipping the black and white colors. Since inscriptions are typically white characters on a black background, flipping ensures better visual clarity. The image is then scaled and resized to 256×256 pixels.

Text matching establishes a one-to-one correspondence between extracted images and the original text, using heuristic rules such as right-to-left and top-to-bottom writing order.

4.1.2 Character Quality Assessment

As historical artifacts, calligraphy works often suffer from quality issues. As a result, some extracted images may be of low quality, with noise or missing or broken strokes. To prevent these from affecting stroke extraction and characterization, we assess image quality using PP-OCR [8] to verify whether the recognition result confidently matches the intended character. We also manually inspect correctly recognized images to filter out those with minor issues, such as moderate noise. After this process, 2,050 instances are retained.

4.1.3 Stroke Extraction

Extracting strokes allows us to characterize characters at finer stroke and local structure levels, enabling a representation that more faithfully reflects the established criteria. We adopt the model by Li et al. [25]. However, some strokes are not fully captured because the training dataset includes only fixed-width strokes, while strokes in calligraphy often vary in width. Following suggestions from the model developers, we increase stroke width by gradually expanding each contour point along its local normal direction. We then make manual adjustments to correct any remaining inconsistencies. After extraction, we annotate horizontal strokes to better support character characterization.

4.2 Character Characterization

We characterize characters based on the criteria "Pinghua Kuanjie" and "Xiehua Jinjie" [38]. The former mainly denotes characters with relatively flat horizontal strokes and ample spacing between strokes, while the latter primarily describes characters with more pronounced slants in horizontal strokes, denser spacing, and little negative space.

The criteria explicitly reflect the perspectives of skewness and compactness, and implicitly capture wholeness and thickness. Wholeness shows how skewness and compactness affect the character's overall shape and ink usage, while thickness complements the stylistic perception conveyed by the criteria, such as dynamics and strength. Accordingly, we analyze characters from these four perspectives, introducing detailed features for each, some of which are illustrated in Fig. 2. Wholeness describes the high-level overall structure of the character. It consists of two features: aspect ratio (**AR**) and ink ratio (**IR**).

- **AR** calculates the ratio of the width of the character portion (black pixel portion) to its height.
- **IR** computes the ratio of the character portion to the total image size (256×256).

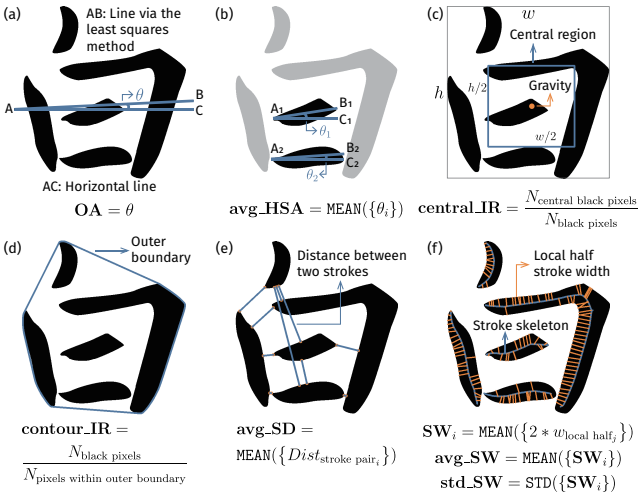


Fig. 2: Illustration of character features. Note that the pixels on the skeleton (f) are sampled for better visual clarity.

Skewness measures the character's slope and its variation. It includes three features: overall angle (**OA**), average horizontal stroke angle (**avg_HSA**), and standard deviation of stroke angles (**std_SA**).

- **OA** measures whether the character appears to rise or remain flat from left to right (Fig. 2a). This is determined by fitting a straight line to the character portion using the least squares method and converting the slope of the line into an angle.
- **avg_HSA** analyzes the inclination of horizontal strokes (Fig. 2b). Each stroke is fitted with a straight line using the least squares method, and its angle is calculated from the slope of the line. The feature is defined as the mean of these angles.
- **std_SA** measures the variation in stroke angles. Similarly, each stroke is fitted with a straight line, and its angle is calculated. The feature is defined as the standard deviation of these angles.

Compactness evaluates the amount of white space within the character. It comprises three features: central ink ratio (**central_IR**), contour ink ratio (**contour_IR**), and average stroke distances (**avg_SD**).

- **central_IR** measures the proportion of ink (black pixels) near the character's center of gravity relative to its total ink amount (Fig. 2c). To calculate this, the character's center of gravity, actual width, height, and total ink amount are determined. Using the center of gravity as the central point, the ink amount within the central region defined by half the character's actual width and height is computed. The value is the ratio of the ink in this region to the total ink amount.
- **contour_IR** measures the ink density within the character's outer boundary (Fig. 2d). The boundary is identified by detecting all contours, merging the contour points, and computing their unified convex hull. The measurement is obtained by dividing the total ink amount by the area of the convex hull.
- **avg_SD** quantifies the spacing between strokes (Fig. 2e). It is calculated by finding the closest distances between the contour points of each stroke pair, summing these distances, and dividing the total by the number of stroke pairs.

Thickness evaluates the width of strokes and its variation. It includes three features: average stroke ink amount (**avg_SIA**), average stroke width (**avg_SW**), and standard deviation of stroke widths (**std_SW**).

- **avg_SIA** is calculated by dividing the total ink amount by the number of strokes.
- **avg_SW** is calculated as follows (Fig. 2f). Each stroke is skeletonized into a single-pixel-wide connected structure that preserves its topology. For each pixel on this skeleton, the distance to the nearest background pixel is computed. This distance represents half of the local stroke width, as the path from skeleton to background is approximately perpendicular to the stroke direction at that point. The full stroke width is obtained by doubling these half-widths and averaging them. Finally, the value is the average width calculated over the entire set of strokes.
- **std_SW** is calculated as the standard deviation of widths across the entire set of strokes. (Fig. 2f).

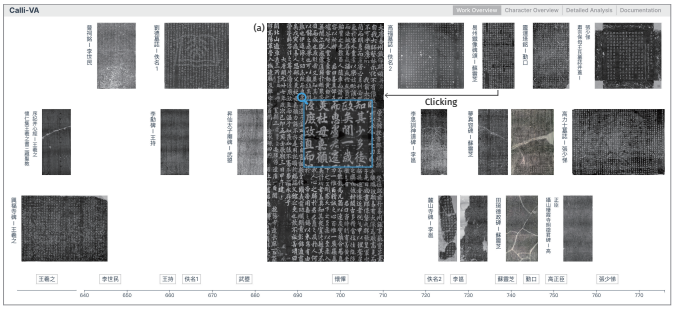


Fig. 3: The work overview part displays inscriptions and their corresponding calligraphers. Users can view an inscription in detail (a).

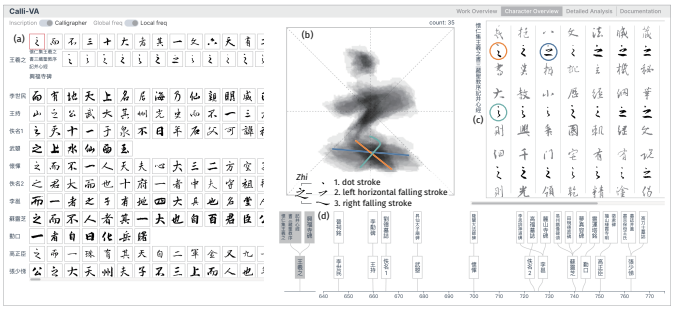


Fig. 4: The character overview part consists of four views: the Character Matrix View (a), the Superimposed Character View (b), the Character Context View (c), and the Timeline View (d).

5 Visual Design

We develop Calli-VA based on the design requirements outlined in Sec. 3.2. The system consists of three components, each aligned with a specific design requirement (Fig. 1). The overview component supports character selection for analysis (**R1**) and includes the work overview (**R1.1**) and the character overview (**R1.2**). The detailed analysis component enables in-depth character analysis (**R2**) and includes three parts: exploration, which presents characterization results (**R2.1**) and allows flexible selection of character instances (**R2.2**), explanation, which helps interpret results (**R2.3**), and recommendation, which assists in adjusting analysis granularities and perspectives (**R2.4**). Finally, the documentation component helps users record their findings (**R3**).

5.1 Overview

The overview component (**R1**) includes both the work overview and the character overview.

5.1.1 Work Overview

The work overview part (Fig. 3) displays inscriptions along with their corresponding calligraphers (**R1.1**). X-positions are mainly based on temporal data, except for two inscriptions from the Collected Characters of Wang Xizhi Series, which are placed at the leftmost to indicate the stylistic origin of the others. Specifically, we calculate the optimal number of rows based on the total image size and screen size. Inscriptions are then distributed across rows to minimize overlap and balance row widths. Finally, x-positions are fine-tuned to resolve any remaining overlaps. Please refer to the appendix for a detailed description.

Calligraphers are positioned based on the average x-positions of their inscriptions. Clicking an inscription opens it in full screen, where a magnifying lens enables detailed exploration of specific regions (Fig. 3a).

5.1.2 Character Overview

The character overview part (Fig. 4) allows users to explore characters from inscriptions and calligraphers (**R1.2**) through four views.

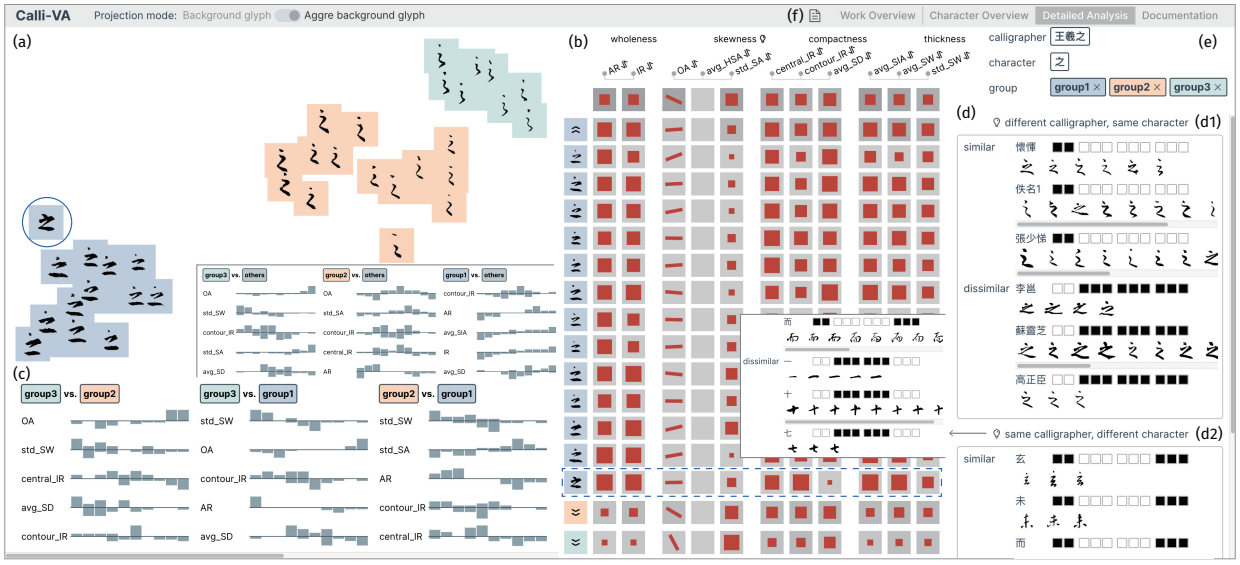


Fig. 5: The detailed analysis component consists of three parts. The exploration part includes the Projection View (a) and the Table View (b). The explanation part is handled by the Pattern Explanation View (c). The recommendation part includes granularity recommendations (d) and perspective recommendations integrated into the Table View header. (e) serves as a legend, and clicking (f) saves the current analysis results.

The Character Matrix View (Fig. 4a) organizes characters from each object into a matrix, with rows representing objects and cells representing characters. Users can toggle between object types and character sorting strategies using buttons at the top. To avoid overwhelming users with numerous instances, each cell contains a representative instance of the character, i.e., the one with the highest average similarity to all other instances. Clicking on a cell displays all related instances, which appear below the cell to avoid shifts in focus. When viewing calligraphers, instances are grouped by their corresponding inscriptions, making it easier to observe possible differences across inscriptions.

Clicking a character also activates the two views on the right. The Superimposed Character View (Fig. 4b) overlays all related instances onto a "mizi" grid, a calligraphy guide that divides the square into eight triangles using four intersecting lines. Instance transparency adjusts automatically based on frequency, highlighting dominant writing styles. The Character Context View (Fig. 4c) displays each instance's neighboring characters within its original inscription, offering insight into the stylistic choices behind the selected character.

The Timeline View (Fig. 4d) complements the part by showing the correspondence between inscriptions and calligraphers, maintaining the same order as in the work overview part. Background highlights indicate the inscriptions and calligraphers associated with the character selected in the Character Matrix View.

When users identify characters of interest, they can double-click to select them and proceed to the detailed analysis component.

5.2 Detailed Analysis

The detailed analysis component (**R2**, Fig. 5) consists of three parts: exploration, explanation, and recommendation.

5.2.1 Exploration

The exploration part (**R2.1** and **R2.2**) has two views. The Projection View (Fig. 5a) visualizes relationships between instances by representing similarity as spatial proximity. We normalize feature vectors obtained through characterization and compute a similarity matrix based on Euclidean distances. For instances without horizontal strokes, we substitute missing **avg_HSA** values with zero, avoiding incomplete relationship representations when analyzing characters with and without horizontal strokes together. The similarity matrix is projected using t-SNE [43], chosen for its effectiveness in identifying local patterns [50]. Instances are displayed as images rather than points for intuitiveness.

Users can group instances using lasso brushing. In addition to objects and characters, instances may also belong to groups. We set the affiliation priority from highest to lowest as group, object, and character. Colors from the Tableau 10 palette [1] encode these affiliations based on priority (Fig. 5e). Specifically, groups are always color-coded, objects are color-coded if there is more than one, and characters are color-coded only when a single object contains multiple characters.

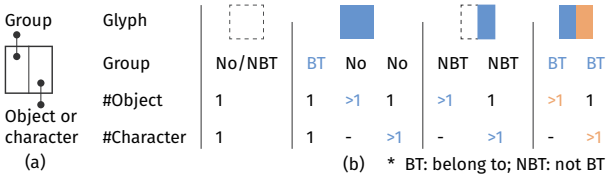


Fig. 6: Design of the background glyph. (a) illustrates the overall design, while (b) demonstrates its variations under different conditions. Colored texts indicate which information is shown in the glyph.

We provide two types of glyphs for representing instance affiliations through color encoding. Background glyphs (Fig. 6) highlight individual instance affiliations and are divided into two sections: the left indicates group affiliation, and the right shows object or character affiliation, depending on which is color-coded. If only one section is color-coded, it fills the entire glyph. In contrast, aggregated background glyphs emphasize shared affiliations among adjacent instances. They display the highest-priority color-coded affiliation and are created by merging the backgrounds of overlapping instances into a unified polygon. Glyph types can be switched using the button at the top.

The Table View (Fig. 5b) presents detailed feature information for instances. The header lists the four perspectives and their associated features. The body contains two types of rows: instance rows, which show data for individual instances, and aggregated rows, which summarize data for subsets of instances based on affiliation priority. Specifically, if groups exist, instances are aggregated by group. Instances without a group are aggregated by object, and if only one object exists, by character. If there is only one object and one character, no aggregation is applied. The first aggregated row always summarizes all instances.

All rows, except the first aggregated row, include a background glyph indicating affiliations. The background glyph design for instance rows matches that in the Projection View (Fig. 6). In aggregated rows, the left section remains the same, given the highest priority of group aggregation. The right section adapts to reflect the presence of multiple objects or characters. Multiple fixed-width rectangles are drawn, with their number corresponding to the count of objects or characters and their heights proportional to relative frequencies.



Fig. 7: Design alternatives for the foreground element in each cell of the Table View, differing in the visual channels utilized.

Each cell displays the corresponding feature value. For the two angle-related features (**OA** and **avg_HSA**), the value is encoded by the angle between a line and the x-axis. Other features are represented by the area of a foreground square. In aggregated rows, average feature values are shown using lines or foreground squares, while variation among instances is indicated by the luminance of background squares.

We considered several visual channels for encoding the foreground element in each cell, including position, length, angle, area, and luminance (Fig. 7). For angle-related features, the angle channel was selected for its intuitive semantic alignment. For other features, angle was excluded to prevent visual confusion. Then, as luminance is already used in aggregated rows, we excluded it for the same reason.

Among the three remaining channels, position and length are traditionally considered the most effective for representing quantitative data [30]. However, we chose the area channel for two key reasons. First, a recent study [31] demonstrates that as the number of visual marks increases, the perceptual precision of position, length, and area becomes relatively comparable. Second, calligraphy scholars prefer the area channel. In their research, they often focus on the structural layout of characters. They find that squares of different areas, as highly symmetrical shapes, align more naturally with their perceptual habits.

Multiple interactions are integrated into the Table View. Users can sort rows based on a specific feature. A hierarchical sorting method is used: the first aggregated row remains fixed, while other aggregated rows, instance rows within each aggregated row, and unaggregated instance rows are sorted independently. Additionally, users can fold or unfold instance rows within an aggregated row by clicking its icon. They can also focus on a specific perspective by clicking its name in the header (**R2.1**), which updates the Projection View accordingly.

5.2.2 Explanation

The Pattern Explanation View (Fig. 5c) helps interpret results from the Projection View in terms of features (**R2.3**). According to Brehmer et al.'s characterization [2], analyzing a projection involves first identifying clusters and then understanding their characteristics. In our case, "clusters" refer to instances within the same affiliation. However, since instances belonging to the same object or character may not form dense regions, we use the term "subsets" instead to avoid confusion. Following the affiliation priority, we only offer explanations for subsets based on the highest-priority existing affiliations.

The goal is to reveal how subsets form and how they differ. Revealing subset formation requires pairwise comparison between a subset and all other instances, while understanding differences between subsets has two strategies: pairwise comparison or multi-subset comparison. Although multi-subset comparison may seem more general, we choose pairwise comparison for two reasons. First, it is inherently simpler and can support both goals without changes to the visual design, reducing cognitive load. Second, multi-subset comparison adds algorithmic complexity. As t-SNE is not inherently interpretable and 11 features are involved, we need additional methods to automatically identify significant features. Existing methods [10,23,57] support only pairwise comparison and are difficult to extend, highlighting this complexity.

Considering these factors, we employ ccPCA [10] to identify significant features. The top five features, ranked by their relative contribution, are visualized using diverging histograms. If users focus on a specific perspective, only features within that perspective are considered, and the number of histograms is reduced accordingly.

5.2.3 Recommendation

This part helps users effectively adjust analysis granularities and perspectives (**R2.4**). The Granularity Recommendation View (Fig. 5d) has two categories: the same characters across different objects (d1) and different characters within the same objects (d2). Both categories includes characters similar to and dissimilar from the current ones, revealing broader commonalities and variations. To generate recommendations, we first identify candidates based on the selected object type, objects, characters, and recommendation category. We then compute the similarity between the current characters and each candidate. The three most similar and the three most dissimilar candidates are recommended in the "similar" and "dissimilar" sections, respectively.

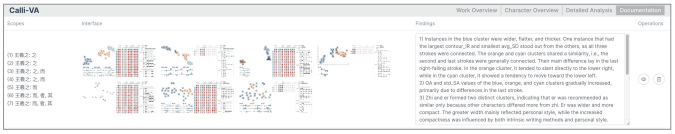


Fig. 8: The documentation component provides information from previous analyses and enable users to record their findings.

We highlight the perspectives contributing significantly to the recommendation. To do this, we calculate the similarity between the current characters and each recommendation separately for each perspective. Perspectives with similarity above the overall similarity are highlighted for similar recommendations, while those below it are highlighted for dissimilar recommendations. To avoid the influence of varying feature counts, normalization is applied. In the view, features are represented as squares and grouped by perspective in the same order as in the Table View. Features from highlighted perspectives are filled in black. If users focus on a specific perspective, the granularity recommendation is restricted to that perspective, and perspective-level highlighting is replaced with feature-level highlighting.

Recommended perspectives are marked with an icon next to their names in the Table View header. Inspired by subspace clustering [19], we recommend perspectives where instances form clusters. We apply HDBSCAN [3], an algorithm capable of detecting clusters with varying densities, to each perspective, then assess the significance of the detected clusters. If only one cluster is found, we assess its intra-cluster distance. Otherwise, we use the Davies-Bouldin Index [7] to evaluate both inter- and intra-cluster distances. A predefined threshold determines whether a perspective is recommended. When groups exist, we evaluate whether they remain distinct clusters using the same methods, focusing on perspectives that highlight their differences.

Besides the three parts, a legend (Fig. 5e) shows existing groups, objects, characters, and their color encodings. Users can directly remove items from the legend to adjust the analysis granularity (**R2.2**).

5.3 Documentation

Once users complete the current analysis, they can save the results by clicking the documentation button (Fig. 5f). They can then switch to a new granularity or perspective, or summarize their findings from previous analyses in the documentation component (**R3**, Fig. 8).

The component uses a table-based layout, where each row represents a round of analysis and includes the following information for each analysis step: the objects and characters involved, along with a screenshot from the detailed analysis component. The screenshot serves as a reference, preserving users' mental map. After selecting it, users can return to the detailed analysis component or delete the step. Findings are recorded in the "Finding" column.

6 Evaluation

We invited both collaborating and independent calligraphy scholars to evaluate the Calli-VA system. The involvement of additional scholars was to minimize bias, particularly the risk of overestimating the system's effectiveness. This section presents three typical cases and their feedback collected from follow-up interviews.

6.1 Case 1: Wang Xizhi's Calligraphic Style

One scholar began by analyzing the two works from the Collected Characters of Wang Xizhi Series, as Wang Xizhi's style influenced all the others. While viewing the inscriptions (Fig. 3), he noticed that the lower-left one was of poor quality, with many unrecognizable characters. Therefore, he focused his analysis on the other inscription.

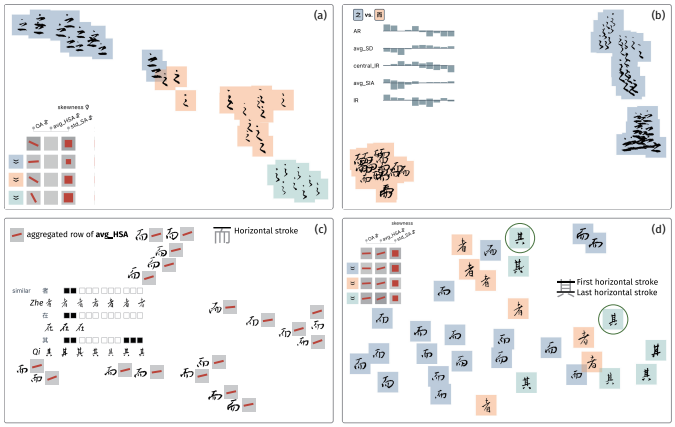


Fig. 9: Analysis of Wang Xizhi's calligraphic style. (a) analyzed *zhi* in the skewness perspective. (b) compared *zhi* and *er*. (c) and (d) analyzed the horizontal strokes of *er*, *zhe*, and *qi*.

After entering the character overview part, he organized the Character Matrix View by calligrapher and local frequency sorting (Fig. 4a). He then noticed that the commonly used character *zhi* (之) appeared most frequently. Clicking the corresponding cell displayed all relevant instances, none of which came from the low-quality inscription. The Superimposed Character View (Fig. 4b) revealed three main styles of *zhi*, differing primarily in the last right falling stroke (indicated by different colors). The Character Context View (Fig. 4c) further showed variations in the second left horizontal falling stroke, i.e., whether it connected to the last stroke. The scholar then selected the character and proceeded to the detailed analysis component.

6.1.1 Analyzing the Character *Zhi*

The scholar observed that the three previously identified styles formed distinct clusters (Fig. 5a). The central cluster was slightly closer to the upper-right cluster, suggesting greater similarity between them. To better analyze the three styles, he created three corresponding groups.

From the Table View (Fig. 5b) and the Pattern Explanation View (Fig. 5c), the scholar observed that instances in the blue cluster were wider (larger **AR**), flatter (smaller **OA**), and thicker (larger **IR** and **avg_SIA**). Besides, one instance (outlined in blue) stood out from the rest. It had the largest **contour_IR** and smallest **avg_SD**, as all three strokes were connected. On the other hand, the orange and cyan clusters shared a similarity: the second and last strokes were generally connected (both had smaller **avg_SD**). Their main difference was in the last right falling stroke, i.e., whether it simply slanted to the lower right or angled further toward the lower left. This was reflected in **OA**—smaller in the orange cluster and larger in the cyan cluster.

Next, the scholar noticed that the skewness perspective was recommended, indicating that the three clusters remained distinct. He switched to this perspective and founded that the **OA** and **std_SA** values of the blue, orange, and cyan clusters gradually increased (Fig. 9a), primarily due to differences in the last stroke.

The analysis above revealed differences among the three writing styles, reflecting that Wang Xizhi developed his own diversified style by drawing on the strengths of many others. Moreover, variations within each style echoed his calligraphic philosophy that "every stroke must contain a certain degree of variation." On the other hand, shared features across the styles—such as their dynamic and graceful qualities, indicated by large **OA**, **std_SA**, or **std_SW**—embodied the elegance and free spirit of the Wei and Jin dynasties in which he lived.

The scholar then decided to analyze Wang's other characters. To this end, he checked the Granularity Recommendation View (Fig. 5d). The three dissimilar characters were expected since they were flat with no stroke spacing. Among the three similar characters, he chose the last one, *er* (而), as it was also a commonly used character.

6.1.2 Comparing the Characters *Zhi* and *Er*

Zhi and *er* formed two distinct clusters (Fig. 9b), indicating that *er* was recommended as similar only because other characters differed more from *zhi*. The Pattern Explanation View showed that *er* was wider (larger **AR**) and more compact (smaller **avg_SD** and larger **central_IR**). The greater width mainly reflected personal style, while the increased compactness was influenced by both intrinsic writing methods and personal style. The first dot stroke of *zhi* naturally had spacing from the others, and in one of his styles, Wang also added spacing between the second and third strokes (as seen in blue-cluster instances in Fig. 5a).

6.1.3 Analyzing the Horizontal Strokes

The scholar then removed *zhi* (Fig. 9c) and focused on the horizontal stroke, finding minimal variation among instances. The **avg_HSA** values were similar, and the background square of the aggregated row appeared light. The Granularity Recommendation View showed that *er* was similar to *zhe* (者) and *qi* (其), both of which contain four horizontal strokes. He selected them to continue his analysis.

Skewness was not a recommended perspective, indicating no significant pattern (Fig. 9d). The Table View showed that their horizontal stroke styles were relatively similar (similar **avg_HSA**). The main difference was that *qi* showed greater variations, as reflected by the darker background square of the aggregated row. The scholar found a compensatory pattern, where the first horizontal stroke was flatter and the last more slanted (two classic examples are outlined in cyan).

6.2 Case 2: From Wang Xizhi to Other Calligraphers

Another scholar reviewed the findings recorded by the first scholar (Fig. 8). She returned to the detailed analysis component displaying *zhi* by Wang and noticed that Zhang Shaoti and Su Lingzhi, two renowned calligraphers, had also written *zhi* many times in similar and distinct styles (Fig. 5d1), respectively. Curious about these patterns, she first added Zhang's *zhi* to the component.

6.2.1 Comparing the Character *Zhi* Among Three Calligraphers

The Projection View (Fig. 10a) showed that the three clusters of Wang's *zhi* were largely preserved, with Zhang's *zhi* falling within two of them. The similarity among instances in the central cluster was clear. However, differences in the upper-right cluster were noticeable: the second and last strokes in Zhang's *zhi* were generally connected.

To understand why these instances still formed a single cluster, the scholar created a group to compare them with other instances. The Pattern Explanation View revealed that they were more compact (larger **central_IR** and **contour_IR**), wider (larger **AR**), flatter (smaller **OA**), and thicker (larger **avg_SW**). These similarities compensated for the differences in their second and third strokes.

Next, the scholar added Su's *zhi* to the analysis (Fig. 10b). The Pattern Explanation View revealed key differences between Wang's and Su's styles: Su's *zhi* were more compact (smaller **avg_SD** and larger **contour_IR** and **central_IR**), wider (larger **AR**), and had greater variations in stroke width (larger **std_SW**). The scholar also observed that a small subset of Su's *zhi*, positioned in the upper center, resembled one of Wang's styles. In addition, the previous upper-right cluster split into three clusters in the lower right. From top to bottom, the first cluster contained both Zhang's and Su's *zhi*, the second was mostly Wang's, and the third was mostly Su's.

To further explore the differences between Wang's and Su's styles, the scholar created two groups based on the clusters primarily composed of their respective *zhi* (Fig. 10c). In addition to earlier observations, he found that Su's *zhi* were more slanted (larger **OA**), thicker (larger **avg_SW**), and had more consistent variations in stroke angle (a more concentrated **std_SA** distribution).

6.2.2 Comparing More Between Wang Xizhi and Su Lingzhi

The scholar then examined Wang's and Su's other characters to see whether their differences remained consistent. For the character *er* (Fig. 10d), the Pattern Explanation View indicated that Su's *er* was narrower (smaller **AR**) and thicker (larger **IR** and **avg_SIA**). Although it had larger **avg_SD**, the scholar still considered it more compact due

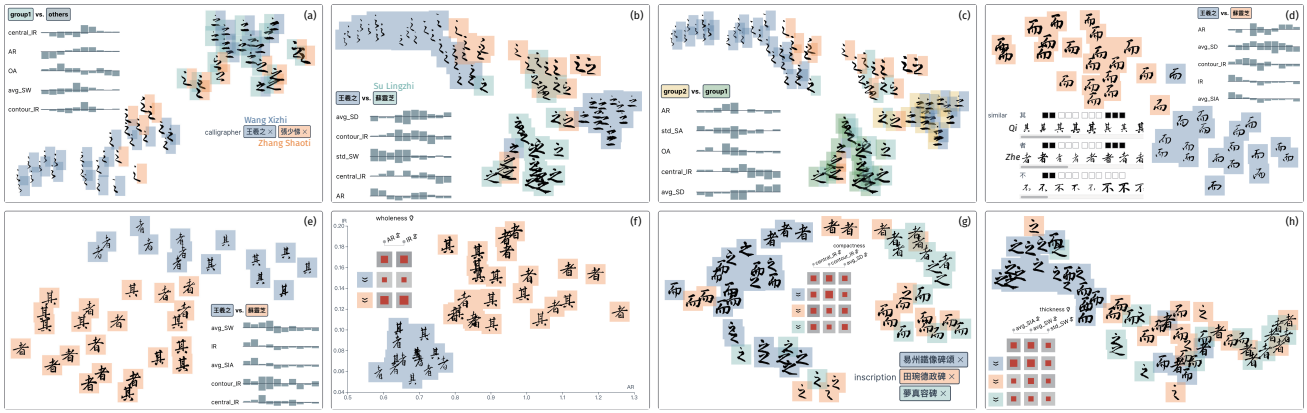


Fig. 10: Analysis of calligraphic styles from Wang Xizhi to others. (a) analyzed *zhi* by Wang and Zhang Shaoti. (b) and (c) added Su Lingzhi's *zhi* to the analysis. (d) compared *er* by Wang and Su, while (e) and (f) compared their *zhe* and *qi*. (g) and (h) compared the three inscriptions by Su.

to its significantly higher **contour_IR**. The narrower style contradicted the earlier finding. In addition, the Granularity Recommendation View suggested that the differences in *zhe* and *qi* were similar to those in *er*, so the scholar continued to analyze these two characters.

These two characters also differed significantly (Fig. 10e). The Pattern Explanation View showed that Su's characters were thicker (larger **avg_SW**, **IR**, and **avg_SIA**) and more compact (larger **contour_IR** and **central_IR**). However, analysis in the recommended wholeness perspective (Fig. 10f) revealed that Su's characters were actually wider (larger **AR**). This suggested that width was not a consistent style, whereas compactness and thickness remained constant.

The analysis above revealed stylistic similarities between Zhang and Wang, as well as clear differences between Su and Wang. Although Zhang lived later than Su, his style was more strongly influenced by Wang. This is supported by Zhao Han, a Ming dynasty epigrapher and bibliophile, who praised Zhang for "deeply inheriting the spirit of Wang's calligraphic style." In contrast, Su lived during the reign of Emperor Xuanzong, who promoted a new aesthetic trend that favored fullness and plumpness. Su studied Xuanzong's style meticulously and absorbed it into his own writing, leading the Song dynasty literatus Wei Tai to describe his work as "so thick as to border on turbidity."

Furthermore, looking at Su's handwriting alone, it is not all of the same degree of compactness and thickness. Since he wrote three inscriptions, the scholar next analyzed the differences among them.

6.2.3 Comparing Three Inscriptions by Su Lingzhi

The scholar chose three frequently occurring characters, *zhi*, *er*, and *zhe*, focusing on the perspectives of compactness and thickness (Fig. 10g and h). She found that instances from the orange and cyan inscriptions were intermixed in both perspectives, indicating stylistic similarity. However, instances from the blue inscription were distributed differently. This inscription was an early work by Su, while the other two were created during his mature period. The Table View showed that instances in his later works had greater variations in stroke width and were less compact and thinner. This reflected Su's stylistic progress: although his writing remained relatively thick and compact, he achieved a more fluid and dynamic effect by moderately reducing the stroke width and increasing spacing between strokes, creating contrast between thickness and thinness, compactness and openness.

6.3 Case 3: More Complex Characters

Previously analyzed characters, such as *zhi* and *er*, were relatively simple. Therefore, one scholar chose to analyze a more complex character. Considering both frequency and complexity, he selected *jun* (君), which appeared 23 times in the works of four calligraphers: Anonymous 1, Anonymous 2, Su Lingzhi, and Zhang Shaoti.

The Projection View (Fig. 11a) showed that each calligrapher's instances distributed roughly together, suggesting greater differences between calligraphers than within them. Instances by Anonymous 2 formed a distinct cluster in the lower right, accompanied by only one

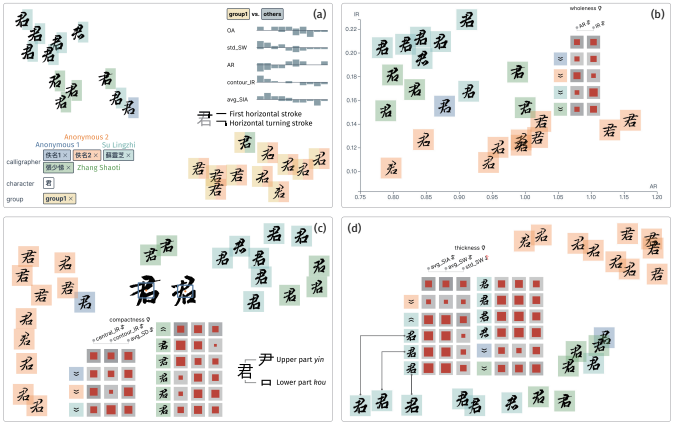


Fig. 11: Analysis of *jun* from (a) all perspectives, and the recommended perspectives: (b) wholeness, (c) compactness, and (d) thickness.

instance from Zhang, while all other instances clustered in the upper left. This indicated Anonymous 2's unique style. The scholar created a group to explore these two clusters further. He found that instances in lower-right cluster were flatter (smaller **OA**), wider (larger **AR**), less compact (smaller **contour_IR**), thinner (smaller **avg_SIA**), and had smaller variations in stroke width (smaller **std_SW**).

The scholar next noted variations within the cluster. In instances on the left, the first horizontal stroke was so long that it crossed the horizontal turning stroke, while in those on the right, it was shorter and left a gap before the turning stroke. This distinction added to the stylistic diversity, e.g., greater width could result from a longer horizontal stroke or overall horizontal stretching. Moreover, the five features from the Pattern Explanation View corresponded to three perspectives, which were recommended when no group was created. The scholar then explored each perspective separately for deeper insights.

From the perspective of wholeness (Fig. 11b), **AR** showed that increased width primarily resulted from a longer horizontal stroke. In all of Anonymous 2's instances, those with the first horizontal stroke crossing the horizontal turning stroke had a larger **AR**, a pattern also seen in Zhang's instances. Even in cases with a shorter first horizontal stroke, Anonymous 2's instances remained wider due to their previously noted less compact style. Meanwhile, **IR** indicated that Su's instances were consistently the thickest.

From the perspective of compactness (Fig. 11c), the Table View indicated that instances by Anonymous 1 and 2 were less compact, whereas those by Su and Zhang were more compact. Additionally, Zhang's instances were divided into two clusters, primarily differing in their **central_IR**. The two instances located in the upper middle had a larger **central_IR** compared to the three in the upper right. This result was somewhat unexpected, since the upper-middle instances featured

a longer first horizontal stroke, typically pushing ink outward. Upon careful examination, the scholar found that these two instances had thicker strokes within the central region, especially at the end of the horizontal turning stroke, and a closer positioning of the upper (*yin*) and lower (*kou*) parts, as shown by a smaller **avg_SD**.

In analyzing thickness (Fig. 11d), Anonymous 2's instances remained distinctly thinner with less variation in stroke width compared to those by other calligraphers. In addition, the scholar observed that Su's instances, which were all from his mature period, largely resembled those of the other two calligraphers, except for three positioned separately in the lower left. These three instances had the smallest **std_SW**. The scholar concluded that, after increasing the variations in stroke width, although these instances still remained thicker than the others, the differences narrowed significantly.

The analysis identified two primary styles for Anonymous 2's *jun*, corresponding respectively to two scripts within the work. Instances with a longer first horizontal stroke crossing the horizontal turning stroke represented the more vigorous regular script, while those with a shorter one reflected the more elegant semi-cursive script. Some scholars have noted stylistic similarities between him and Wang Xizhi, supported by Zhang (recognized as inheriting Wang's stylistic essence) also using both styles of *jun*. Furthermore, the analysis highlighted differences between Anonymous 2's and Zhang's styles, supporting the view that Anonymous 2 was also influenced by the subtle and unadorned style of early Tang dynasty calligrapher Yu Shinan.

6.4 Expert Feedback

We conducted one-on-one interviews with scholars after they used the system and summarized their feedback as follows.

Analysis workflow. All scholars agreed that the structured workflow significantly improved the efficiency of their analysis. One scholar noted, "I fully appreciate the power of visual analytics. It seamlessly combines automated analysis techniques with interactive visualizations, allowing me to use techniques I am not familiar with while also applying my own expertise." Another scholar commented, "I can easily identify characters for analysis through the overviews. Besides presenting the characterization results, the system helps explain patterns I am interested in and recommends other potential analysis scenarios, making the analysis process much more smooth."

Learning curve and usability. The system may appear complex at first glance, as it consists of three components, each comprising one or two complete interfaces. However, scholars did not find it difficult to learn or use, primarily for two reasons. First, the system is well-organized according to the design requirements they endorse. Each component corresponds to a high-level design requirement and is further logically divided based on lower-level ones. Second, the views and interactions are intuitive. Even before the system was introduced, scholars could infer the functions of most views on their own. The only view they initially found challenging was the Projection View, but once we explained that "similarity $\approx$ proximity," they quickly grasped it.

Visual design and interactions. Overall, scholars found the visual design satisfactory and felt the interactions supported their needs. However, they also identified issues affecting their experience. The main design concern was that switching perspectives or granularities sometimes made instance positions in the Projection View hard to track. This occurred because the system did not account for previous positions when re-running t-SNE. Inspired by the incremental dimensionality reduction method [9], this issue could be mitigated by geometric transformations. Regarding interactions, scholars noted the inconvenience of switching to the documentation component to record their findings. Although a screenshot of the relevant analysis was included, they suggested it would be more convenient to record insights directly through a pop-up panel within the detailed analysis component.

7 Discussion

This section discusses the limitations and possible extensions.

Generalizability. We discuss the generalizability of our system across three areas: semi-automatic processing, calligraphy style analysis, and broader analysis scenarios.

First, all three steps preceding character characterization (Sec. 4.1) begin with automated methods, followed by manual intervention to ensure high quality while reducing human effort. This semi-automatic approach can effectively leverage human expertise to achieve better results when fully automated methods fall short. Similar strategies are widely used in data cleaning [12], classification [21], and annotation [63]. Therefore, this approach is broadly generalizable.

Second, calligraphy works have different sources and script styles. Beyond inscriptions, calligraphers also write on paper. The key difference is that inscriptions feature white characters on a black background, while papers have black characters on a white or yellow background. The processing steps are nearly identical, except that paper-based characters do not require color flipping due to their natural visual clarity.

This paper focuses on the semi-cursive script. There are also four other main scripts: seal, clerical, regular, and cursive [35]. Except from cursive script, where characters often flow together and strokes are frequently modified or omitted, the other three generally feature neatly structured characters and strokes. Thus, our analytical approach of characterizing individual characters and analyzing them from overview to detail remains applicable. Note that characterization criteria may need adjustment to better suit the aesthetic view of a specific script.

Finally, our workflow can be abstracted into three steps: selecting items, analyzing them, and recording findings. This aligns with the information-seeking mantra [40]. While alternatives exist, such as detail first [27], this mantra is widely adopted in visual design, suggesting that our workflow is broadly generalizable. In the more relevant humanities field, for example, it can be applied to analyze stylistic evolution in elements of paintings or patterns on clothing.

Scalability. Computations are categorized as global or local. Global computations, such as character and stroke extraction and character characterization (Sec. 4), are pre-computed and do not affect system loading time. Conversely, local computations, which mainly handle selected character instances within the detailed analysis component (Sec. 5.2), are performed on the fly. As the number of instances is relatively small (typically a dozen to a hundred), the system runs smoothly.

Regarding visual scalability, several features help manage complexity and reduce clutter: manual preprocessing interfaces, the magnifying lens in the work overview part, and instance aggregation in the Character Matrix View and the Table View. However, as the number of works and character instances grows, all three areas need to be further improved. First, in manual preprocessing, users currently process instances one by one without considering relationships between them. Instead, presenting similar instances in sequence, such as those of the same character, can significantly reduce time and cognitive effort [33]. Second, while the lens in the work overview part aids in browsing individual works, exploration could be more efficient with added guidance, such as indicating where characters of interest are located. Besides, the connection between the work overview part and the character overview part can be strengthened to create a more seamless experience. For example, the system could automatically highlight relevant works or characters based on user interactions in the work overview part. Third, in displaying character instances, the Projection View may suffer from visual occlusion because instances are shown as images rather than points. This could be mitigated by clustering similar instances and displaying representative examples from each cluster.

8 Conclusion

This paper presents Calli-VA, a visual analytics system that offers full-process support for analyzing and comparing Chinese calligraphic styles. After character and stroke extraction and character characterization, Calli-VA supports analysis through three components: overview, detailed analysis, and documentation. Explanation and recommendation mechanisms are also integrated to help users understand patterns and guide further exploration. Three case studies and expert feedback demonstrate the system's usefulness and effectiveness. In the future, we aim to improve visual scalability in manual preprocessing, work overview exploration, and character instance aggregation.

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